

ARTIFICIAL INTELLIGENCE EAI404

E-content (NOTES- UNIT 3,4,5)

Unit-3: Physical Symbol Systems (8 Hours)

Physical Symbol System Hypothesis

A Physical Symbol System is the foundation of Artificial Intelligence.

Definition:

A physical symbol system is a system that uses symbols to represent knowledge and performs reasoning by manipulating these symbols.

Key Idea:

AI systems think by:

- Representing facts using symbols
- Applying rules to those symbols
- Producing intelligent behavior

Example:

Humans use language symbols, AI uses logic symbols.

Topics in Unit-3

1. Propositional Logic

Definition:

Propositional Logic is the simplest form of logic where statements are either:

✓ True or False

◆ Propositions:

A proposition is a declarative sentence.

Example:

- P: "It is raining"
- Q: "The road is wet"

◆ Logical Operators:

Operator	Meaning	Example
AND ($\wedge$)	Both true	$P \wedge Q$
OR ($\vee$)	Either true	$P \vee Q$
NOT ($\neg$)	Negation	$\neg P$
IMPLIES ($\rightarrow$)	If-Then	$P \rightarrow Q$

2. First Order Logic (FOL)

📌 Definition:

First Order Logic is more powerful than propositional logic because it includes:

- Objects
- Predicates
- Quantifiers

◆ Example:

[
Human(Socrates)
]

Meaning: Socrates is human.

Quantifiers:

Symbol	Meaning	Example
$\forall$	For all	$\forall x \text{ Human}(x)$
$\exists$	There exists	$\exists x \text{ Doctor}(x)$

FOL is widely used in AI for knowledge representation.

3. Forward Chaining

Definition:

Forward chaining is a reasoning method that starts from known facts and applies rules to reach conclusions.

Process:

Facts → Rules → New Facts → Goal

Example:

1. Fact: It is raining
2. Rule: If raining → road wet
3. Conclusion: Road is wet

Used in expert systems

4. Backward Chaining

Definition:

Backward chaining starts from the goal and works backward to check if facts support it.

Process:

Goal → Rules → Required facts → Check

Example:

Goal: Road is wet

Rule: If raining → wet

Check: Is it raining?

✓ Used in diagnosis systems

5. Resolution

📌 Definition:

Resolution is a rule of inference used for proving statements logically.

It works by:

- Converting logic into clauses
- Applying resolution rule
- Deriving contradiction or proof

Example:

1. $(P \vee Q)$

2. $(\neg P)$

Resolution gives $\rightarrow Q$

Resolution is important in automated theorem proving.

6. Probabilistic Reasoning

📌 Definition:

Probabilistic reasoning deals with uncertainty in AI.

Because real-world information is not always certain.

Example:

"There is a 70% chance it will rain."

Uses:

- Bayes Theorem
- Machine learning prediction

- Medical diagnosis

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

7. Utility Theory

Definition:

Utility theory helps AI choose the best action based on usefulness or benefit.

Example:

Self-driving car decision:

- Stop (high safety utility)
- Speed (low safety utility)

AI selects action with maximum utility.

$$\text{Expected Utility} = \text{Probability} \times \text{Reward}$$

Used in:

- ✓ Decision making
- ✓ Reinforcement Learning
- ✓ Game AI

Summary Table

Concept	Meaning	Use in AI
Propositional Logic	Simple true/false logic	Basic reasoning
First Order Logic	Logic with objects & quantifiers	Knowledge representation
Forward Chaining	Data-driven reasoning	Expert systems

Concept	Meaning	Use in AI
Backward Chaining	Goal-driven reasoning	Diagnosis systems
Resolution	Proof technique	Automated reasoning
Probabilistic Reasoning	Handles uncertainty	Prediction & inference
Utility Theory	Decision based on reward	RL and planning

✅ Unit-4: Knowledge Representation (8 Hours)

Knowledge Representation (KR)

📌 Definition:

Knowledge Representation is a method used in AI to store information about the world in a form that a computer can understand and reason with.

◆ Goal:

To represent knowledge so that AI systems can:

- ✓ Understand facts
 - ✓ Make decisions
 - ✓ Solve problems
 - ✓ Learn new concepts
-

✅ Topics in Unit-4

1. Semantic Networks (Semantic Nets)

📌 Definition:

A Semantic Network is a graphical way to represent knowledge using:

- Nodes → Objects or concepts
- Links → Relationships between them

Example:

Dog → is-a → Animal

Dog → has → Tail

Features:

- ✓ Easy to understand
- ✓ Shows relationships clearly

Uses:

- Natural Language Processing
 - Expert systems
 - Concept representation
-

2. Frames

Definition:

Frames are structured data representations used to store information about an object in the form of slots.

Structure:

Frame: Car

Color: Red

Wheels: 4

Engine: Petrol

Key Points:

- Frames represent stereotyped knowledge
- Slots can have default values

Uses:

- ✓ Object representation
 - ✓ Knowledge storage in expert systems
-

3. Scripts

Definition:

Scripts represent knowledge about common events or situations in a sequence.

Example: Restaurant Script

1. Enter restaurant
2. Order food

3. Eat food
4. Pay bill
5. Leave

Scripts help AI understand routine activities.

Uses:

- Story understanding
 - NLP applications
 - Event prediction
-

4. Conceptual Dependency (CD)

 **Definition:**

Conceptual Dependency is a representation method that shows the meaning of sentences using primitive actions.

Example:

Sentence: "Ram gave a book to Sita"

Conceptual form:

- Actor: Ram
- Action: Give
- Object: Book
- Receiver: Sita

Purpose:

- ✓ Removes language ambiguity
- ✓ Represents meaning, not words

Uses:

- Natural language understanding
- Machine translation

✓ Learning in Knowledge Representation

5. Learning Automation

📌 Definition:

Learning automation means AI systems improve performance automatically by learning from experience.

Example:

Spam filter learning from emails.

6. Discovery Learning

📌 Definition:

Discovery learning is when AI finds new patterns or knowledge from data without being explicitly told.

Example:

Finding hidden trends in medical datasets.

7. Analogy Learning

📌 Definition:

Analogy learning means solving a new problem using similarity with an old known problem.

Example:

- Solar system model → Atom structure model

Used in reasoning and creativity.

8. Genetic Algorithms (GA)

📌 Definition:

Genetic Algorithms are search and optimization techniques inspired by natural evolution.

Working Steps:

1. Selection
2. Crossover
3. Mutation
4. Best solution evolves

Uses:

- ✓ Feature selection
- ✓ Hyperparameter tuning
- ✓ Optimization problems

✓ Summary Table

KR Method	Representation Style	Use
Semantic Nets	Graph nodes & links	Relationship modeling
Frames	Slot-filler structure	Object description
Scripts	Event sequences	Routine activity understanding
Conceptual Dependency	Meaning-based structure	NLP understanding
Automation Learning	Self improvement	Smart systems
Discovery Learning	Pattern finding	Data mining
Analogy Learning	Similarity reasoning	Problem solving
Genetic Algorithms	Evolution-based optimization	ML tuning

Unit-5: PROLOG & LISP.

Unit-5: PROLOG & LISP (8 Hours)

Unit-5 focuses on AI Programming Languages, especially:

- PROLOG (Logic Programming)
- LISP (Functional & Symbolic Programming)

These languages are widely used in Artificial Intelligence because they support:

- ✓ Symbolic reasoning
 - ✓ Knowledge representation
 - ✓ Problem solving
-

✓ 1. PROLOG (Programming in Logic)

✚ Introduction

PROLOG stands for:

```
[  
PROgramming\ in\ LOGic  
]
```

It is based on First Order Predicate Logic.

Key Idea:

Instead of writing step-by-step procedures, we write:

- **Facts**
 - **Rules**
 - **Queries**
-

✓ PROLOG Syntax

Facts

Facts represent true statements.

Example:

human(ram).

male(ram).

Meaning: Ram is human and male.

Rules

Rules define relationships.

Example:

mortal(X) :- human(X).

Meaning: If X is human, then X is mortal.

Queries

Queries ask questions.

Example:

?- mortal(ram).

Output:

Yes

 Numeric Functions in PROLOG

Prolog supports arithmetic operations:

X is 5 + 3.

Result:

X = 8

Other operators:

- + Addition
- - Subtraction

- * Multiplication
 - / Division
-

✓ List Manipulation in PROLOG

Lists are very important in AI.

Example:

List = [1,2,3,4].

Head and Tail

[H|T] = [1,2,3].

Output:

- H = 1
 - T = [2,3]
-

✓ Predicates

A predicate represents properties or relations.

Example:

likes(ram, mango).

Meaning: Ram likes mango.

✓ Conditionals

Prolog uses :- for conditions.

Example:

happy(X) :- rich(X).

Meaning: X is happy if X is rich.

✓ Recursion in PROLOG

Recursion is common for loops.

Example: Factorial

fact(0,1).

fact(N,F) :-

N > 0,

N1 is N-1,

fact(N1,F1),

F is N*F1.

✓ Input / Output in PROLOG

Example:

write('Hello AI').

nl.

- write prints output
 - nl moves to next line
-
-

✓ 2. LISP (List Processing)

✚ Introduction

LISP stands for:

[
LISt\ Processing
]

It is one of the oldest AI languages, designed for symbolic computation.

✓ Features of LISP

- ✓ Based on functions
 - ✓ Uses lists as main structure
 - ✓ Excellent for AI research
-

✓ Basic Syntax in LISP

Everything is written inside parentheses:

Example:

```
(+ 5 3)
```

Output:

8

✓ List Manipulation in LISP

Example list:

```
(setq mylist '(1 2 3 4))
```

Important Functions:

Function Meaning

CAR First element

CDR Remaining list

CONS Add element

Example:

```
(car '(1 2 3)) → 1
```

```
(cdr '(1 2 3)) → (2 3)
```

```
(cons 0 '(1 2)) → (0 1 2)
```

✓ Recursion in LISP

Example: Factorial


```
(defun fact (n)
  (if (= n 0)
      1
      (* n (fact (- n 1)))))
```

✔ 3. Other AI Programming Languages

Apart from PROLOG and LISP, AI also uses:

Language Use

Python Machine Learning, Deep Learning

Java Expert Systems, NLP tools

C++ Robotics, high-performance AI

R Statistical AI and Data Science

Julia AI optimization and ML

✔ Summary Table

Feature	PROLOG	LISP
Paradigm	Logic Programming Functional Programming	
Focus	Facts & Rules	Symbolic Computation
Main Structure	Predicates	Lists
Best for	Expert systems	AI research & NLP
